artificially produced radioactive isotope of beryllium. Capturing a planetary electron, it changes into the lithium nucleus $_3Li^7$. The formula expressing this process is:

$$_4Be^7 + _{-1}e^0 \rightarrow _3Li^7 + _0\nu^0$$

Such processes are not so very rare. They are frequently associated with fairly long half-lives. The half-life of $_4Be^7$ is approximately fifty-three days. In our Table IV, the atomic nuclei which change by *K*-capture are indicated by circles.

And finally, there is still another process, discovered by Hahn and Strassmann in 1938. This is the reaction already mentioned in which nuclei split into two parts of approximately equal size. Occasionally it may occur spontaneously, too. But this process will be discussed under the heading of artificially induced nuclear transmutations.

IV. ARTIFICIALLY INDUCED NUCLEAR TRANSMUTATIONS

As was first demonstrated by Rutherford, a nuclear transmutation can be produced artificially by shooting some particle into the nucleus. In the majority of the cases, this particle will remain in the nucleus without producing any reaction. But the result may be that a particle of some kind is emitted in turn by the nucleus. If this emitted particle is not of the same kind as the first particle, this reaction means a nuclear transmutation.

Bohr formulated the following theory about such a reaction: When a particle is hurled at the nucleus and actually strikes it, it usually remains in it, for the simple reason that it is held fast there by the exceedingly powerful nuclear forces. As a result, the energy of this particle becomes distributed very rapidly among the other particles in the nucleus, and eventually throughout the entire nucleus. Figure 24 shows a schematic picture of a nucleus. A neutron is approaching it from the outside. The white and black circles represent the nuclear neutrons and protons. As shown by the arrows, these nuclear particles receive an impact from the foreign neutron and, in turn, collide with other particles. When the neutron has penetrated into the interior of the nucleus and its energy has become distributed among all the particles within it, the situation can be expressed in the simplest form as follows: The atomic nucleus is being heated. A pile of sand becomes heated in